

Discharge behavior of Li-Mg-B alloy/MnO₂ couples with LiNO₃-KNO₃-Mg(OH)NO₃ eutectic electrolyte



Yongqiang Niu^{a,b}, Zhu Wu^{a,*}, Junlin Du^{a,b}, Chaohui Pu^a

^a Research Center for New Energy Technology, Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, Shanghai 200050, China

^b University of Chinese Academy of Sciences, Beijing 100049, China

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ABSTRACT

A novel LiNO₃-KNO₃-Mg(OH)NO₃ ternary molten salt system is used as electrolyte with Li-Mg-B alloy/MnO₂ couples for possible use as a power source for geothermal borehole applications. Chemical-compatibility studies of these materials are conducted using simultaneous differential thermal analysis (DTA) and thermo gravimetric analysis (TGA). The ionic conductivity of the new electrolyte has been evaluated by electrochemical impedance spectroscopy (EIS) and its value range from 0.107 to 0.457 S cm⁻¹ in the temperature range from 150 to 300 °C. The discharge behavior of the system is studied over a temperature range of 150 to 300 °C at current densities from 10 to 30 mA cm⁻². At 200 to 300 °C, discharge behavior seems unaffected by discharge rate. However, the system has some specific restrictions associated with temperature.

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1. Introduction

Thermal batteries are non-rechargeable power sources, which use electrolytes of inorganic salts that are solid and considered non-conducting at ambient temperatures. Upon ignition of an internal pyrotechnic heat source, the electrolyte melts and becomes conductive, thereby providing power to an external load. Thermal batteries are used in applications that require extremely long shelf-life and a burst of power from milliseconds to a few hours. The Li(Si)/FeS₂ and Li(Si)/CoS₂ couples with halide-electrolyte system have been extensively studied and used in thermal batteries for nuclear weapons and for missile applications [1]. The typically operating temperature of thermal batteries is between 350 °C and 550 °C.

In recent years, there is an interest in the development of a high-temperature power source that can be used to power data logging instrumentation while drilling in boreholes for oil/gas exploration and geothermal exploration. Operating temperature can range from 200 to well over 400 °C in such an environment. The temperature range is above the thermal window of normal ambient-temperature Li battery, but below that of standard

thermal battery. To meet the requirements of wide range of borehole conditions, current thermal battery technology needs to be modified, or alternate suitable technology allowing the batteries to function well under ambient heat in deep boreholes requires to be developed. One possible approach with modification of the current thermal battery technology is presented in this paper to meet the borehole requirements.

Currently, modified Li-Mg/SOCl₂ cells (manufactured by Battery Engineering, Inc. (Canton, MA)) are used for powering instrumentation used in deep boreholes involved in oil and gas exploration. However, these cells are rated for operation at 180 °C [2]. The temperature limitation of these cells requires that they must be insulated from the immediate thermal environment by the use of a metallic vacuum Dewar. Sandia is initially involved in eliminating the expensive Dewar and internal pyrotechnic and using the heat of the borehole to maintain the electrolyte in molten state [3]. An electrolyte having an even lower melting point is to be selected in oil and gas boreholes, where temperatures reach only 250 °C or so [2].

The highest electromotive force (emf) of thermal batteries can be realized by using pure Li as an anode material. However, since the melting point of lithium is 181 °C, it would be liquid at temperatures of borehole conditions. The so-called liquid anode or 'LAN' was conceived by CRC using Fe power to act as a binder [4]. This requires a large amount of Fe power (70 to 85%), which greatly reduces the potential specific energy and energy density of the anode. (Unless

* Corresponding author. No.235, Chengbei Road, Jiading District, Shanghai, China
Tel.: +86 21 69976861; fax: +86 21 69976863.

E-mail address: wuzhu@mail.sim.ac.cn (Z. Wu).

otherwise noted, all compositions are reported in weight percent.) This type of anode was successfully used in a variety of batteries [5].

The use of a lithium–boron alloy, developed by the Naval Surface Warfare Center, has a great potential because it contains elemental Li within the pores of the Li–B structure [6]. McManis has investigated lithium–boron anodes in nitrate thermal battery cells in the temperature range of 270 to 350 °C, and his studies shows that no appreciable anode polarization occurred in molten nitrate electrolyte at 300 mA cm^{−2} [7]. The use of Mg as an addition to Li–B alloys to produce ternary alloys has also been conceived [8], but electrochemical tests have rarely been conducted.

In an attempt to extend the liquidus region to lower temperatures, several eutectic molten nitrates were evaluated for possible borehole use. The LiNO₃–KNO₃ eutectic electrolyte, which melts at 124.5 °C and has an ionic conductivity of 0.875 S·cm^{−1} at 287 °C, have been studied in several systems [9]. In addition, this electrolyte is stable at temperatures above 400 °C [10]. The use of high-activity anodes with nitrate electrolyte is possible only due to the formation of a protective passive oxide film on the anode; much in the same way that a film of LiCl prevents continues reaction of the Li anode in Li/SOCl₂ cells [11]. The conventional sulfides are not thermodynamically or kinetically stable with molten nitrates and violently react [12], so that oxides must be used instead.

Many studies on the basic electrochemistry of various materials used with the LiNO₃–KNO₃ eutectic electrolyte have been conducted by Miles [13]. Giwa has investigated the Li(Al)/Ag₂CrO₄ couples in the LiNO₃–KNO₃ eutectic electrolyte, but only over a narrow temperature range of 160 to 215 °C and at current densities from 10 to 100 mA cm^{−2} [14]. Guidotti has examined the Li(Al)/MnO₂ couples with LiNO₃–KNO₃ eutectic electrolyte over a temperature range of 150 to 300 °C at current densities from 7.6 to 30.4 mA cm^{−2} [15]. The use of Mg(NO₃)₂ as an addition to LiNO₃–KNO₃ eutectic electrolyte to produce ternary eutectic electrolytes has not yet been reported.

A number of transition metal oxides are suitable for use with molten nitrates [16]. Using MnO₂ as a cathode in molten nitrate electrolyte system is appealing because it is understood to be more environmentally friendly [17]. MnO₂ is chemically compatible with molten nitrate electrolytes at temperatures well over 300 °C [12]. Discharge characteristics of Li–Mg–B Alloy/MnO₂ couples in molten LiNO₃–KNO₃–Ca(NO₃)₂ ternary electrolyte has been studied [18]. This system can be activated at 150 °C and operated over a temperature range of 150 °C–300 °C to produce open circuit voltages of 3.1–3.4 V and initial operating voltages above 2.90 V at 10 mA cm^{−2}. However, both temperature and current density hugely affect cell capacity.

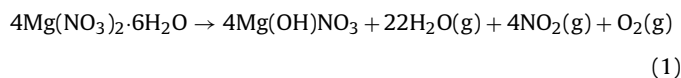
This paper will focus on the galvanostatic discharge tests of Li–Mg–B alloy/MnO₂ couples in LiNO₃–KNO₃–Mg(OH)NO₃ (25.5%, 61.3% and 13.2%, respectively) ternary eutectic electrolyte. These materials are tested in single cells over a temperature range of 150 to 300 °C at current densities from 10 to 30 mA cm^{−2}.

2. Experimental

2.1. Materials

The molten salt mixtures are prepared from LiNO₃, KNO₃ and Mg(NO₃)₂·hexahydrate. The compounds LiNO₃ (purity ≥99%) is bought from Shanghai Fengshun Chemical Reagent Co. (Shanghai, China). The compounds KNO₃ (purity ≥99%) is bought from Shanghai Lingfeng Chemical Reagent Co. (Shanghai, China). The compounds Mg(NO₃)₂·hexahydrate (purity ≥99%) is bought from Sinopharm Chemical Reagent Co. (Shanghai, China).

The required quantities of components amounts of the individual salts are weighed into alumina crucible and dissolved in deionized water. Then the alumina crucible is heated in furnace open to the atmosphere. The temperature is initially limited to approximately 100 °C to allow the water of hydration of the solution to evolve slowly. When the temperature is above the melting point of magnesium nitrate hexahydrate (95 °C), dehydration of magnesium nitrate hexahydrate generates magnesium hydroxide nitrate which is due to reaction (1):



If the reaction is not completed, loss of capacity and drop in voltage will be observed caused by reaction (2) when cells are discharged.



After visual indications of vapor evolution ceased, the melt is heated to 300 °C, and maintained at 300 °C for over 16 hours, followed by quenching and grinding (100 mesh). Then the electrolyte powder is stored in sealed containers.

The separator mixture is prepared by blending the electrolyte powder with 35% MgO, then fused again at 300 °C for over 16 hours, followed by quenching and grinding (100 mesh). The cathode is made up of 70% MnO₂, 10% graphite (as a conductive additive) and 20% electrolyte. The anode is Li–Mg–B alloy (58%, 4% and 38% respectively, before prepared). After prepared, the Li–Mg–B alloy contains 55% Li, 3.8% Mg and 41.2% B. The theoretical capacity of the Li–Mg–B alloy is 1016.0 mAh g^{−1} [18].

2.2. Thermal Analyses

To determine materials compatibilities, thermal analyses of select materials are carried out under high-purity argon (<1 ppm each of water and oxygen) using simultaneous differential thermal analysis (DTA) and thermo gravimetric analysis (TGA) (STA 449 F3 Jupiter simultaneous thermal analyzer, Netzsch; Boston, MA, USA). These samples are sealed in Al₂O₃ pans and heated from 40 to 500 °C at rate of 10 K min^{−1}.

2.3. Electrochemical measurements

The ionic conductivity is determined by electrochemical impedance spectroscopy (EIS) in the temperature range of 150 to 300 °C. Each experiment is repeated at least five times, and an average value is given. The reference solution is 20% NaCl solution at 26 °C. The experiment is carried out with a CHI660D (Chenghua; Shanghai, China) electrochemical workstation including frequency response analysis modules for impedance measurements over the frequency range of 100 KHz to 1 Hz.

2.4. Single-cell Testing

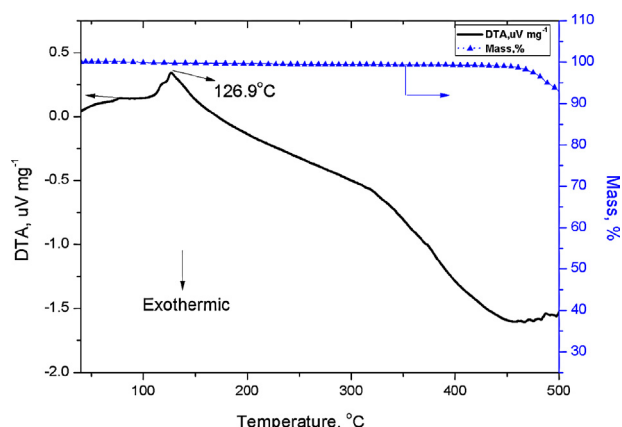
The cathode and separator mixture are cold-pressed into a 15.5 mm diameter pellet under an applied pressure of 296 MPa. The cathode mass is 0.2 g, and the separator mass is 0.2 g. The anode pellets, cut off from the Li–Mg–B alloy sheet, are 16 mm diameter, 0.40 mm thick and the average weight is 0.09 g. The 304 stainless steel is compatible with the molten nitrate, but both Fe and Mo reacted exothermically at 222 °C and 334 °C [12], respectively. The 304 stainless steel pellets (20 mm diameter and 0.1 mm thick) are used as current collectors.

The amount of materials in each part (anode, separator, and cathode) is summarized in Table 1. The galvanostatic discharge

Table 1

The masse percentage of materials in each part (anode, separator, and cathode).

Materials	Li-Mg-B alloy	MnO ₂	Graphite	MgO	Electrolyte
Anode	100%	0	0	0	0
Separator	0	0	0	35%	65%
Cathode	0	70%	10%	0	20%

**Fig. 1.** Simultaneous DTA/TGA responses of the LiNO₃-KNO₃-Mg(OH)NO₃ eutectic electrolyte when heated in argon at 10 K min⁻¹.

behavior of single cells is conducted by a CT2001A battery test system (LAND; Wuhan, China) over a temperature range of 150 to 300 °C in a glove box under an atmosphere of high-purity argon (<1 ppm each of water and oxygen). Steady-state loads of 10, 20 and 30 mA cm⁻² are used. The cell discharge is terminated when the voltage dropped below 0 V.

3. Results and Discussion

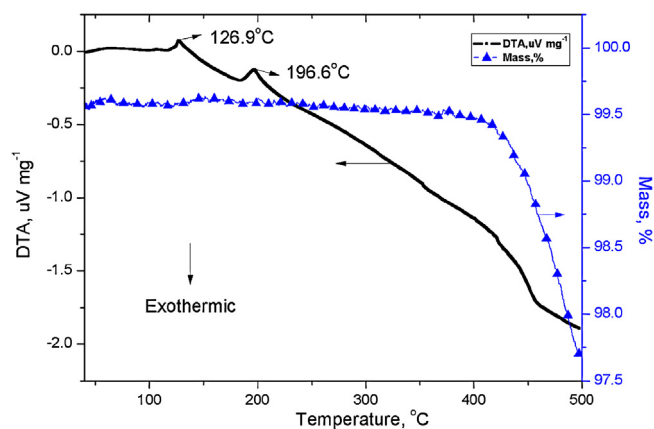
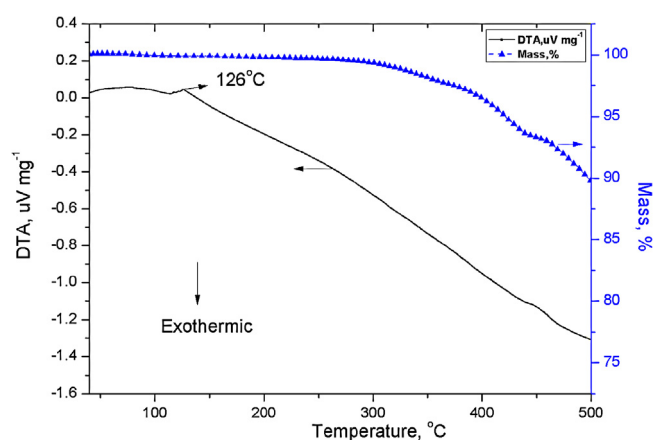
3.1. Thermal Analyses

Regardless of the final chemistry of the system used in the borehole environment, these materials' compatibility must be addressed. Since these materials must be stable for the proposed environment over the operating temperature, the initial part of the study focuses on using thermal analysis as a diagnostic tool.

Simultaneous DTA/TGA examinations of LiNO₃-KNO₃-Mg(OH)NO₃ ternary eutectic electrolyte are shown in Fig. 1. The onset of an endotherm near 111.7 °C, followed by the major endotherm at 126.9 °C, is caused by the melting of the electrolyte. Besides, one onset temperature of losing weight around 450 °C is due to the thermal decomposition of the ternary nitrate electrolyte with producing NO₂ gas.

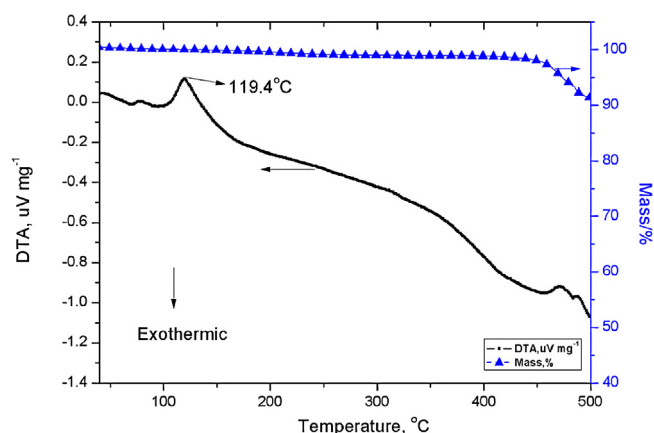
Simultaneous DTA/TGA tests of the Li-Mg-B alloy anode in contact with the LiNO₃-KNO₃-Mg(OH)NO₃ ternary eutectic electrolyte are shown in Fig. 2. From this figure, we find two onset temperatures for endothermic events. The onset of the first endotherm around 117.2 °C, followed by the major endotherm at 126.9 °C, is due to the melting of the electrolyte. Another endotherm onset begins around 187 °C, followed by the major endotherm at 196.6 °C, and occurs because of the melting of lithium in the Li-Mg-B alloy [18]. One onset temperature of losing weight around 400 °C is caused by the thermal decomposition of the electrolyte with producing NO₂ gas under the effect of the Li-Mg-B alloy.

Similar DTA/TGA traces for the full cathode mixture which contains 70% MnO₂, 10% graphite and 20% electrolyte, are shown in Fig. 3. One endotherm is observed for the full cathode mixture. The onset of the endotherm around 111.9 °C, followed by the major endotherm at 126 °C, is resulted from the melting of the electrolyte.

**Fig. 2.** Simultaneous DTA/TGA responses of the Li-Mg-B alloy with the LiNO₃-KNO₃-Mg(OH)NO₃ eutectic electrolyte when heated in argon at 10 K min⁻¹.**Fig. 3.** Simultaneous DTA/TGA responses of the cathode when heated in argon at 10 K min⁻¹.

Besides, we find two onset temperatures of losing weight. The onset of losing weight around 270 °C is due to the oxidation-reduction reaction between graphite and MnO₂ with producing CO gas under the action of the ternary nitrate electrolyte. Another onset of losing weight around 450 °C is caused by the thermal decomposition of the ternary nitrate electrolyte with producing NO₂ gas.

The compatibility of 304 stainless steel pellet with the novel electrolyte is measured by simultaneous DTA/TGA tests, and results are shown in Fig. 4. One endotherm is observed from Fig. 4. The

**Fig. 4.** Simultaneous DTA/TGA responses of 304 stainless steel pellet with the LiNO₃-KNO₃-Mg(OH)NO₃ eutectic electrolyte when heated in argon at 10 K min⁻¹.

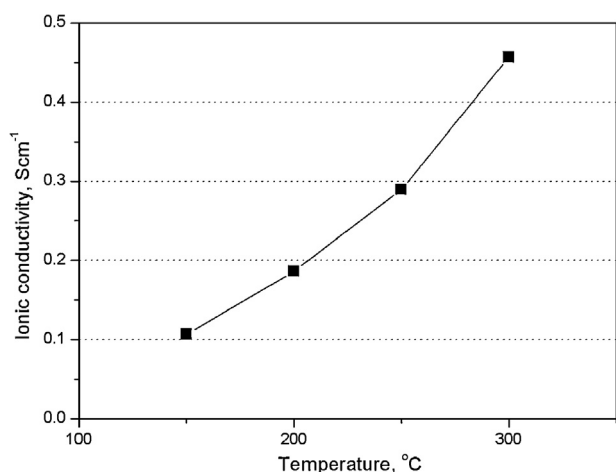


Fig. 5. Evolution of the ionic conductivity versus the temperature.

onset of the endotherm around 108.3°C, followed by the major endotherm at 119.4°C, is resulted from the melting of the electrolyte. One onset temperature of losing weight around 430°C is caused by the thermal decomposition of the electrolyte.

3.2. Ionic conduction behavior

The ionic conductivities of the novel electrolyte are measured by EIS, ranging from 0.107 to 0.457 Scm⁻¹. Fig. 5 shows the relationship between ionic conductivities of the electrolyte and temperatures. The conductivity increases dramatically with the rising of temperature, which is caused by enhanced kinetics at higher temperature.

3.3. Single-cell Tests

Discharge behavior of Li-Mg-B alloy/LiNO₃-KNO₃-Mg(OH)NO₃/MnO₂ single cells at a constant current density of 10 mAcm⁻² is shown in Fig. 6 as a function of temperature. These cells can be activated at 150°C and operated to produce open circuit voltages of 3.2–3.6 V. Three distinct discharge plateaus (2.85 V, 2.60 V, and 1.70 V) are generally evident during discharge, with a weak fourth plateau shown by the cell discharged at 200°C. Based on the relationship between the discharge time and capacity,

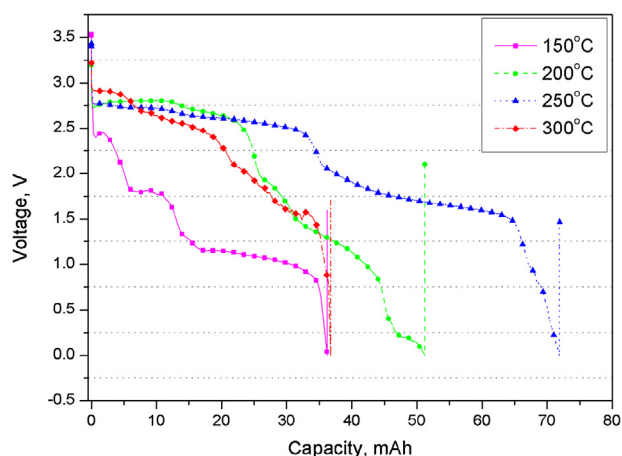
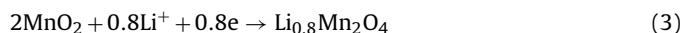


Fig. 6. Discharge behavior of Li-Mg-B alloy/LiNO₃-KNO₃-Mg(OH)NO₃/MnO₂ single cells at 10 mAcm⁻² as a function of temperature.

we speculate that three distinct discharge plateaus are caused by the following reactions.

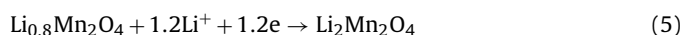
At the first plateau, the cathode reaction is



and the anode reaction is



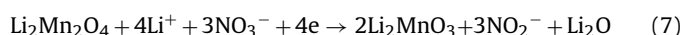
At the second plateau, the cathode reaction is



and the anode reaction is



At the third plateau, the cathode reaction is

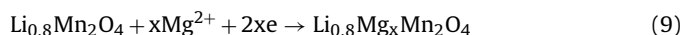


and the anode reaction is



The above reactions have been previously proposed for discharge of Li-Mg-B alloy/MnO₂ couples in molten LiNO₃-KNO₃-Ca(NO₃)₂ eutectic electrolyte [18]. The different potentials of two systems are caused by the different ionic conductivities of two electrolytes.

The fourth plateau is believed to be caused by reaction (9) and reaction (10):



(where x is a trace variable)



Besides, an unwanted reaction may occur during discharging:



This reaction obviously affects the capacity and potential of the system.

Temperature has a hugely effect on the discharge behavior. At 150°C, the lower conductivity of the electrolyte resulted in a relatively low capacity and low potential. The best overall behavior is realized at 250°C, where maxima discharge capacity and better discharge behavior are achieved. When the temperature is up to 300°C, a higher potential is observed caused by the higher ionic conductivity of the electrolyte at higher temperature; however,

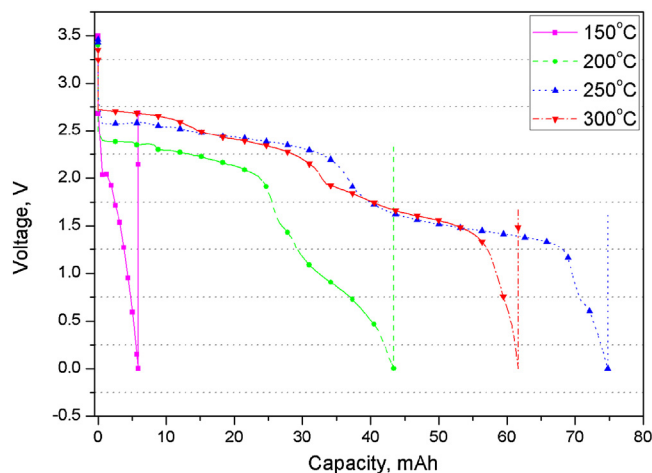


Fig. 7. Discharge behavior of Li-Mg-B alloy/LiNO₃-KNO₃-Mg(OH)NO₃/MnO₂ single cells at 20 mAcm⁻² as a function of temperature.

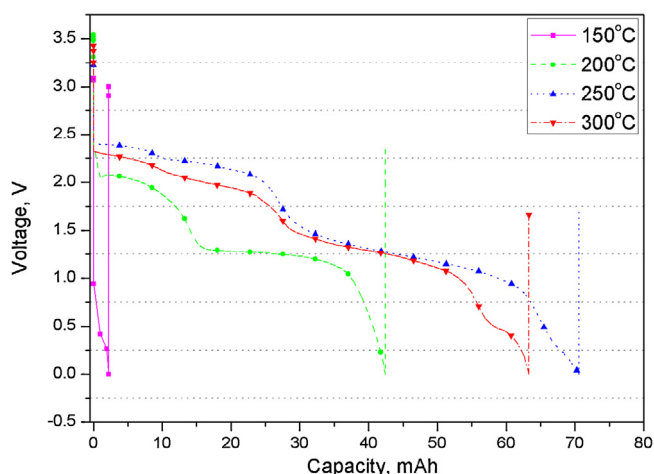


Fig. 8. Discharge behavior of Li-Mg-B alloy/LiNO₃-KNO₃-Mg(OH)NO₃/MnO₂ single cells at 30 mA cm⁻² as a function of temperature.

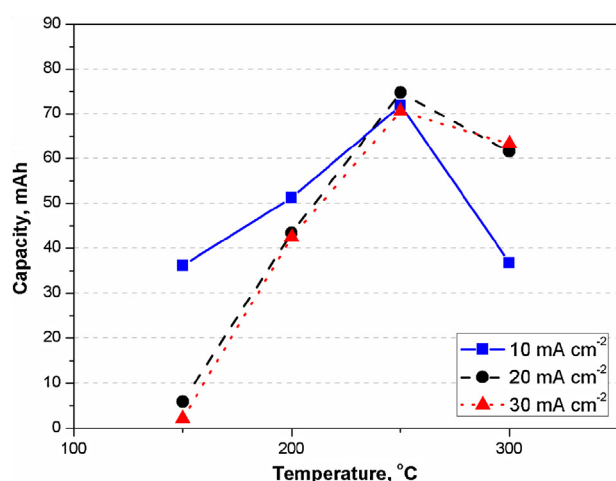


Fig. 9. Effect of temperature and current density on the discharge capacities of Li-Mg-B alloy/LiNO₃-KNO₃-Mg(OH)NO₃/MnO₂ single cells discharged at 10, 20, and 30 mA cm⁻².

a significant decrease in capacity is observed and only two discharge plateaus are evident. This is indicative of increased self discharge at higher temperatures, which is consistent with the observed chemical reaction noted by simultaneous DTA/TGA (Fig. 3) in this region.

Discharge behavior of Li-Mg-B alloy/LiNO₃-KNO₃-Mg(OH)NO₃/MnO₂ single cells at 20 mA cm⁻² is shown in Fig. 7 as a function of temperature. Similar data for current density of 30 mA cm⁻² is presented in Fig. 8. Degraded discharge behaviors (large drop in voltage, big loss in capacity and less voltage plateaus) are observed when these cells are discharged at a higher current density at 150 °C. However, discharge behaviors seem unaffected by discharge rate at 200 to 300 °C, which is caused by the higher ionic conductivity of the electrolyte at higher temperature.

The capacity data are summarized in Fig. 9. There is a competition between loss of capacity by self discharge and increase in capacity caused by enhanced kinetics at the higher temperatures. The discharge capacity of these single cells shows a maximum value near 250 °C, followed by a decrease as the temperature increases.

At 200 to 300 °C, the discharge capacity seems unaffected by discharge rate when these single cells are discharge at 10 to 30 mA cm⁻².

4. Conclusions

The novel LiNO₃-KNO₃-Mg(OH)NO₃ ternary eutectic electrolyte can be used as electrolyte in Li-Mg-B alloy/MnO₂ couples. The ternary eutectic electrolyte melts around 111.7 °C, and the thermal decomposition of the electrolyte occurs around 450 °C. The electrolyte is compatible with Li-Mg-B alloy anode for temperatures between 40 °C and 400 °C. The oxidation-reduction reaction between graphite and MnO₂ begins around 270 °C with producing CO gas under the action of the electrolyte. The ionic conductivity of the electrolyte is determined by EIS and its value range from 0.107 to 0.457 S cm⁻¹ in the temperature range from 150 to 300 °C.

The Li-Mg-B alloy/LiNO₃-KNO₃-Mg(OH)NO₃/MnO₂ thermal battery cells can be activated at 150 °C and operated over a temperature range of 150 to 300 °C to produce open circuit voltages of 3.2–3.6 V. Three distinct discharge plateaus (2.85 V, 2.60 V, and 1.70 V) are generally evident when these cells are discharge at 10 mA cm⁻². At 150 °C, degraded discharge behaviors are observed when these cells are discharged at a higher current density. At 200 to 300 °C, discharge behaviors seem unaffected by discharge rate. However, The Li-Mg-B alloy/LiNO₃-KNO₃-Mg(OH)NO₃/MnO₂ system has some specific restrictions associated with temperature.

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